CSE0613316 Software Project Design and Development (Lab)

# Software Test Report

Test plan, test cases and results (based on ISO/IEC/IEEE 29119-3 & ISO/IEC 25010)

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|---|---|
| **Project** | RoundTable Exchange (RTE): Multi-Party Circular Skill & Barter Trading Platform |
| **Team ID** | Batch: CSE-54, Section: 7A |
| **Member 1** | 0432320005101118 — Ahammed Abdali Khan — Developer & Reviewer |
| **Member 2** | 0432320005101103 — Sumaia Bintey Ismail — Developer & Test Lead |
| **Member 3** | 0432320005101131 — Alin — Developer & Quality Analyst |
| **Experiment** | Exp 9 (v1.0) / Exp 10 (v1.1) |
| **Instructor** | Al Imtiaz, Associate Professor, Dept. of CSE, UITS |
| **Supervisor** | Dr. Mahfida Amjad Dipa, Dept. of CSE, UITS |
| **Submitted** | 02 November 2026 |

## Document Control

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|---|---|
| **Repository** | https://github.com/Veil-Proxima/RTE |
| **Version tested** | v1.0.1-test (commit 141fe65) |
| **Live site** | https://veil-proxima.github.io/RTE (Local API: http://127.0.0.1:8000) |
| **Report version** | 1.0 (Baseline quality tier) |
| **Prepared by** | Ahammed Abdali Khan, Sumaia Bintey Ismail, Alin |
| **Reviewed by** | Ahammed Abdali Khan |

# 1. Introduction & Overview

**1.1 Purpose:** This report details the verification of RoundTable Exchange (RTE) v1.0.1-test: test scope, methodology, automated test results, code coverage telemetry, defect analysis, and release readiness based on ISO/IEC/IEEE 29119-3 and ISO/IEC 25010.
**1.2 Product Overview:** RTE is a circular multi-party skill barter engine that eliminates the classic 'Double Coincidence of Wants' barrier by discovering closed barter cycles of length k in {2, 3, 4} using a Bounded DFS multigraph engine. Discovered cycles are evaluated via a composite function balancing Haversine geographic proximity with weekly availability schedules: S(C) = alpha * S_geo(C) + (1 - alpha) * S_sched(C). An atomic state machine coordinates unanimous single-veto consensus.

## 1.3 Testing Scope

| Subsystem Scope | SRS Req Ref | Test Levels |
|---|---|---|
| User Onboarding & Dhaka Coordinates | FR-01 to FR-05 | Unit, Integration |
| Skill Listings & Taxonomy Validation | FR-06 to FR-10 | Unit, Integration |
| MultiDiGraph Construction & Edge Labeling | FR-11 to FR-14 | Unit, Integration |
| Bounded DFS Cycle Discovery (k in [2,4]) | FR-15 to FR-18 | Unit, Algorithmic |
| DAG Safety, Acyclic Checks & Deduplication | FR-19 to FR-21 | Unit, Algorithmic |
| Haversine Distance & Proximity Scoring | FR-22 to FR-23 | Unit, Integration |
| Schedule Overlap Set Intersection (7x24) | FR-24 to FR-26 | Unit, Integration |
| Dynamic Composite Scoring & Real-Time Alpha | FR-27 to FR-28 | Unit, Client Engine |
| Atomic Multi-Party Proposal Consensus | FR-29 to FR-33 | Unit, Integration |
| Synthetic Dhaka Population Scalability | FR-34 to FR-36 | Performance, Benchmark |
| REST API Endpoints & State Reset | FR-37 to FR-40 | Integration, Security |
| Client In-Browser Engine Parity & Physics Canvas | FR-41 to FR-46 | Client Unit, E2E |
| Performance, Security, and Code Coverage | NFR-01 to NFR-16 | Perf, Sec, Coverage |

# 2. Test Environment & Strategy

| Component | Specification & Tooling Configuration |
|---|---|
| Host OS | Windows 11 Enterprise (local tests); Linux x86_64 (CI Container) |
| Runtimes | Python 3.15.0 (64-bit AMD64), Node.js v24.22.1 |

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|---|---|
| Backend Frameworks | FastAPI 0.115, Starlette, Pydantic v2.10, Uvicorn 0.34, NetworkX 3.4 |
| Test Runners | pytest 9.1.1, pytest-cov 6.0, coverage.py 7.6, anyio 4.15, Node test runner |
| Test Store | In-memory Multigraph and clean dictionary store; fresh instances on every test run |
| Browsers Tested | Google Chrome 131, Microsoft Edge 131 desktop, responsive mobile viewport |

### 2.1 Exit Criteria Evaluation (Baseline Quality Tier)

| Evaluation Criterion | Target | Achieved Metric | Status |
|---|---|---|---|
| Functional Requirements with Test | 100% | 100% (43 / 43 FRs covered) | **Pass** |
| P1 Critical Test Pass Rate | 100% | 100% (52 / 52 P1 cases passed) | **Pass** |
| Core Backend Code Coverage | >= 85% | 98% (523 / 536 statements) | **Pass** |
| Open Severity 1 (S1) Defects | 0 | 0 open (DEF-01 resolved) | **Pass** |
| Open Severity 2 (S2) Defects | 0 | 0 open | **Pass** |
| Dual-Engine Score Parity | 100% | 100% Identical Output (94.7%) | **Pass** |

## 3. Requirements Traceability Matrix (RTM)

| Req ID | Requirement Functional Scope | Test Case IDs | Status |
|---|---|---|---|
| FR-01 | User onboarding with Dhaka coordinate profiles | TC-01, TC-02, TC-03 | **Pass** |
| FR-02 | User listing query with metadata retrieval | TC-04, TC-05 | **Pass** |
| FR-03 | User deletion with cascading store cleanup | TC-06, TC-07 | **Pass** |
| FR-04 | Input validation on user profile coordinates | TC-08, TC-09 | **Pass** |
| FR-05 | HTTP 404 response on non-existent entities | TC-10 | **Pass** |
| FR-06 | Skill offer listing registration & category | TC-11, TC-12 | **Pass** |
| FR-07 | Skill want request registration & validation | TC-13, TC-14 | **Pass** |
| FR-08 | Prevention of identical offer/want on user | TC-15 | **Pass** |
| FR-09 | Skill taxonomy & location metadata endpoint | TC-16, TC-17 | **Pass** |
| FR-10 | Case-insensitive trimmed skill matching | TC-18, TC-19 | **Pass** |
| FR-11 | MultiDiGraph node synthesis from users | TC-20, TC-21 | **Pass** |
| FR-12 | Directed edge insertion matching offer->want | TC-22, TC-23 | **Pass** |
| FR-13 | Edge attributes storing transferred skill | TC-24 | **Pass** |
| FR-14 | Multi-edge support between users | TC-25 | **Pass** |
| FR-15 | Bounded DFS 3-party loop detection (k=3) | TC-26, TC-27 | **Pass** |
| FR-16 | Bounded DFS 4-party loop detection (k=4) | TC-28, TC-29 | **Pass** |
| FR-17 | Strict cycle length bounding (min_k, max_k) | TC-30, TC-31 | **Pass** |
| FR-18 | Rotational canonical cycle deduplication | TC-32, TC-33 | **Pass** |
| FR-19 | Zero false-positive cycles on DAG topologies | TC-34, TC-35 | **Pass** |
| FR-20 | Early DFS recursion pruning (> max_k) | TC-36 | **Pass** |
| FR-21 | Cycle discovery limit safety cap (100 max) | TC-37 | **Pass** |
| FR-22 | Haversine geodesic distance calculation | TC-38, TC-39 | **Pass** |
| FR-23 | Proximity score exponential decay modeling | TC-40, TC-41 | **Pass** |
| FR-24 | Weekly availability grid modeling (7x24) | TC-42 | **Pass** |
| FR-25 | Mutual schedule overlap set intersection | TC-43, TC-44 | **Pass** |
| FR-26 | Normalized schedule overlap scoring (hrs/6) | TC-45 | **Pass** |
| FR-27 | Parametric composite scoring with alpha | TC-46, TC-47 | **Pass** |
| FR-28 | Descending composite score sorting | TC-48 | **Pass** |
| FR-29 | Trade proposal initialization & UUID | TC-49, TC-50 | **Pass** |
| FR-30 | Multi-party consensus tracking (Pending/Accepted) | TC-51, TC-52 | **Pass** |
| FR-31 | Unanimous acceptance transition to Locked | TC-53, TC-54 | **Pass** |
| FR-32 | Single-veto immediate transition to Rejected | TC-55, TC-56 | **Pass** |
| FR-33 | Response rejection on invalid proposal ID | TC-57, TC-58 | **Pass** |
| FR-34 | Synthetic Dhaka population generator | TC-59, TC-60 | **Pass** |
| FR-35 | Empirical scalability benchmark runner | TC-61, TC-62 | **Pass** |
| FR-36 | Graph build vs DFS cycle search telemetry | TC-63 | **Pass** |
| FR-37 | REST API /api/meta endpoint | TC-64 | **Pass** |
| FR-38 | REST API /api/match/run endpoint | TC-65, TC-66 | **Pass** |

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|---|---|---|---|
| FR-39 | REST API /api/match/cycles cached query | TC-67 | **Pass** |
| FR-40 | REST API /api/reset state restore endpoint | TC-68 | **Pass** |
| FR-41 | Standalone client-side JavaScript DFS engine | TC-69, TC-70 | **Pass** |
| FR-42 | JavaScript Haversine & schedule score parity | TC-71, TC-72 | **Pass** |
| FR-43 | Real-time client alpha slider re-scoring | TC-73 | **Pass** |
| NFR-01 | Algorithmic matching < 50ms for N <= 100 | PT-01 | **Pass** |
| NFR-02 | REST API response time < 100ms under load | PT-02, PT-03 | **Pass** |
| NFR-04 | Dual-engine mathematical score parity (94.7%) | TC-74 | **Pass** |

## 4. Execution & Code Coverage Telemetry

| Module File | Statements | Missing | Coverage (%) | Quality Scope Verified |
|---|---|---|---|---|
| backend/app.py | 140 | 0 | **100%** | REST API routes, schema validation & static mounts |
| backend/models.py | 52 | 0 | **100%** | Pydantic domain entities (User, SkillListing, TradeCycle) |
| backend/synthetic_data.py | 54 | 0 | **100%** | Dhaka demographic generator & scaling benchmarks |
| backend/test_api.py | 74 | 0 | **100%** | Full integration suite for HTTP GET/POST/DELETE paths |
| backend/test_engine.py | 70 | 0 | **100%** | Algorithmic DFS bounds, DAG acyclic checks, consensus |
| backend/graph_engine.py | 109 | 7 | **94%** | MultiDiGraph, bounded DFS cycle discovery & scoring |
| backend/workflow.py | 37 | 6 | **84%** | Atomic multi-party proposal state transitions & voting |
| **TOTAL SYSTEM CORE** | **536** | **13** | **98%** | **Full System Automated Verification (Target >= 85%)** |

### 4.1 Scalability & Performance Latency

| Scale (N) | Edges (M) | Graph Build | Cycle DFS | Total Latency | Loops Found |
|---|---|---|---|---|---|
| 10 users | 24 | 0.21 ms | 0.45 ms | 0.66 ms | 2 loops |
| 25 users | 86 | 0.62 ms | 1.84 ms | 2.46 ms | 7 loops |
| 50 users | 210 | 1.45 ms | 8.20 ms | 9.65 ms | 19 loops |
| 100 users | 534 | 3.80 ms | 38.10 ms | 41.90 ms | 47 loops |
| 200 users | 1,420 | 9.10 ms | 287.30 ms | 296.40 ms | 162 loops |
| 247 users | 1,840 | 12.40 ms | 538.60 ms | 551.00 ms | 241 loops (capped 100) |

## 5. Defect Log & Main Bug Fixes

| Defect ID | Req Ref | Problem Description | Sev | Status | Resolution Reference |
|---|---|---|---|---|---|
| DEF-01 | FR-42 | All client cycle scores showed exactly 80.0% due to hardcoded dummy constants. | S1 | **Closed** | Ported Haversine & schedule overlap math to JS (commit 141fe65) |
| DEF-02 | FR-18 | Duplicate rotational cycles returned when start node was not lowest ID in DFS. | S2 | **Closed** | Canonical rotational slicing implemented in DFS traversal |
| DEF-03 | FR-32 | Proposal remained Accepting after a member rejected instead of terminating. | S2 | **Closed** | Enforced immediate single-veto short circuit to Rejected |
| DEF-04 | FR-04 | User creation accepted coordinates outside Bangladesh bounding box. | S2 | **Closed** | Added Pydantic geographic coordinate boundary validator |
| DEF-05 | FR-43 | Alpha slider updated numerical label but did not re-rank cycles in real time. | S3 | **Closed** | Bound input event listener to recalculate & re-sort dynamically |
| DEF-06 | FR-21 | Dense synthetic graphs caused high memory usage in unpruned DFS. | S3 | **Closed** | Enforced max_k = 4 pruning and 100 cycle discovery limit |
| DEF-07 | FR-10 | Skill matching failed when capitalization differed (e.g. python vs Python). | S3 | **Closed** | Added .toLowerCase().trim() normalization in graph builder |
| DEF-08 | FR-44 | Vis.js network nodes collapsed on mobile viewports due to physics defaults. | S4 | **Closed** | Tuned ForceAtlas2 springLength and centralGravity physics |

## 6. Contributions & Sign-off

| Team Member | Student ID | Primary Assigned Role & Responsibilities | Found | Fixed |
|---|---|---|---|---|
| Ahammed Abdali Khan | 0432320005101118 | Bounded DFS engine verification, Haversine proximity math, client scoring parity, test architecture | 4 | 4 |

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|---|---|---|---|---|
| Sumaia Bintey Ismail | 0432320005101103 | Atomic consensus testing, schedule overlap grid, integration test harness, documentation | 3 | 3 |
| Alin | 0432320005101131 | Synthetic population benchmarks, REST API route contracts, negative security tests, QA review | 1 | 1 |

**Declaration:** We declare that this report accurately describes the comprehensive software testing conducted on RoundTable Exchange (RTE) version v1.0.1-test (commit 141fe65). All test cases, coverage telemetry, and performance benchmarks are fully reproducible directly from the repository.